MILITARY SPECIFICATION

CASE, CARTRIDGE, 105MM, M14B4

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not covered by the drawings and provides quality assurance provisions for the fabrication of parts and packing for one type of cartridge case, M14B4.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-C-46487 - Cleaning, Preparation and Organic Coating of Steel Cartridge Cases

STANDARDS

MILITARY

MIL-STD-1167 - Ammunition Data Card
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

FSC: 1315

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

7548320 - Cartons, Ammunition for Interplant Shipment of Brass, Aluminum and Sprially Wrapped Steel Cartridge Cases, Empty, 75MM to 120MM Inclusive
8647259 - Chamber Gage
8595386 - Case, Cartridge, 105MM, M14B4

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI-B46.1 Surface Texture

(Application for copies should be addressed to the American National Standards Institute, (ANSI), 1430 Broadway, New York, NY 10018).

ASTM - American Society for Testing and Materials
B117 - Salt Spray (Fog) Test
E18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Test for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Phila, Pa 19103).

3. REQUIREMENTS

3.1 General.-The cartridge case shall comply with the requirements of the applicable drawings, referenced specifications and the following requirements.

3.2 Protective coating.-Protective coating as specified on the applicable drawings and referenced specifications shall apply to all surfaces of the cartridge case regardless of contour configuration and shall be uniform and continuous.

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3.3 Material.-The supplier shall be responsible for selecting from the classification list contained in the raw material specification the surface and physical condition of the raw material intended for use in manufacture.

3.4 Mechanical properties.-The mechanical properties specified on the applicable drawing and referenced specifications shall be uniform throughout each section.

3.5 Maximum profile and alignment.-Maximum profile and alignment of the cartridge case shall comply with the requirements of the applicable drawing.

3.6 First article inspection.-This specification contains technical provisions for First Article Inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.7 Functional.-The cartridge case shall be subjected to a functional test in the following phases:

Excess Pressure Phase
Cold Temperature Phase
Excess, Excess Pressure Phase (For Information Only)

3.7.1 Obturation.-The cartridge case shall effectively seal the breech of the weapon against leakage of gases generated during firing.

3.7.2 Soundness.-After firing, the cartridge case shall be free of cracks or splits except as permitted by Table II.

3.7.3 Ejection.-After firing, the cartridge case shall be ejected by the service extractors when the weapon is operated in the normal manner (see 6.2).

3.7.4 Primer retention.-The cartridge case shall retain its primer during firing and ejection.

3.8 Workmanship

3.8.1 General.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.8.2 Metal defects.-Prior to firing, the cartridge case shall be free from cracks, blisters, seams, laminations, deep scratches and porosity.

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3.8.3 Body/Base/Collar interface.-There shall be no evidence of interference between the collar and body in the region beyond the body seat groove.

3.8.4 Burr.-No cartridge case shall have a burr.

3.8.5 Foreign matter.-No cartridge case shall contain chips, dirt, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or herein, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections.-The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission.-The contractor shall submit a First Article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The First Article sample shall consist of the following items in sample quantities as indicated.

PART DESCRIPTION	DRAWINGS	QUANTITY
Cartridge Case, Coated	8595386-1-2 or -3 as applicable	270
Cartridge Case, Components, Coated and Unassembled	8595386-1-2 or -3	4
Base, Uncoated	8595383 as applicable	10
Base, Uncoated	11820670 as applicable	10
Base, Uncoated	11751170 as applicable	10
Collar, Uncoated	8595384	10
Test Panels	See MIL-C-46487	

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250 of the above coated cartridge cases shall be forwarded to a designated proving ground for ballistic testing. The sample shall be manufactured using the same materials, equipment, processes, and procedures as will be used in regular production. All parts and materials, including packaging shall be the same as used in regular production and shall be obtained from the same source of supply. A first article sample or portion thereof, as directed by the procuring agency, shall also be submitted whenever a change occurs in manufacturing processes, materials used, drawings, specification, or source of supply, such as to significantly affect product uniformity as determined by the responsible government technical agency.

4.3.2 Inspections to be performed.-The first article parts submitted in accordance with paragraph 4.3.1 will be subjected by the Government to any or all of the examinations and tests specified in the Item Detail Specification and to any or all the requirements of the applicable drawings and to the inspection of Table I.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
	Case, Cartridge, 105MM, M14B4		See Below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cartridge Case, Coated (Dwg. 8595386-1-2 or -3 as applicable)				
	Examination for defects	270		3.1	4.4.4
	Examination of Body/Base/Collar interface	2		3.8.3	4.5.5
	Ballistic test				
	a. Excess, Excess Pressure (Infor Only)	100		3.7	4.5.4
	b. Excess Pressure	100		3.7	4.5.4
	c. Cold Temperature	50		3.7	4.5.4
	Cartridge Case Components, Coated and Unassembled (Dwg. 8595386-1-2 or -3)				
	Thickness of coating	4		3.1	4.5.3
	Salt Spray	2		3.1	4.5.3.1
	Base, Uncoated (Dwg. 8595383, 11751170, 11820670, as applicable)				
	Examination for defects	10		3.1	4.4.4.2
	Hardness	10		3.4	4.5.1

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48423A (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	Case, Cartridge, 105MM, M14B4				8595386
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Collar, Uncoated (Dwg. 8595384 as applicable)				
	Examination for defects	10		3.1	4.4.4.1
	Hardness	10		3.4	4.5.1
	Test Panels (See MIL-C-46487)				
	Weight of coating	2		3.1	See MIL-C-46487
	Flexibility test	2		3.1	See MIL-C-46487

NOTES:

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4.3.3 Rejection.-See MIL-A-48078.

4.4 Quality conformance inspection

4.4.1 Inspection lot formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination.-See MIL-A-48078.

Unless otherwise specified in the classification of defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

4.4.3 Tool control dimensions.-Dimensions marked tool control shall be gaged during First Article testing and after each 250 items produced on each machine in opertion and prior to any tooling change. Where destruction of components is necessary to inspect these dimensions measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool. Corrective action will immediately be instituted by the contractor whenever a defective tool control dimension is found. All material produced prior to periodic inspection will be rejected and screened whenever a defective tool control dimension is found.

4.4.4. Classification of defects

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48423A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.4.1	Collar, Prior to Coating (as applicable)		8595384
			NEXT HIGHER ASSEMBLY 8595386

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	True position of large outside diameter to small outside diameter		0.40%	3.1	Gage
102	Small outside diameter		0.40%	3.1	Gage
103	Distance from rear face to flange, max.		0.40%	3.1	Gage
104	Total length		0.40%	3.1	Gage
105	Distance from rear face to forward inside angle		0.40%	3.1	Gage
106	Thickness of flange		0.40%	3.1	Gage
107	Width of rear surface		0.40%	3.1	Gage
108	Depth of groove		0.40%	3.1	Gage
109	Width of groove wall		0.40%	3.1	Gage
110	Metal defective		0.40%	3.1	Visual
111	Crack in weld zone (if applicable)		0.40%	3.1	Visual
Minor					
201	Large outside diameter		0.65%	3.1	Gage
202	Angle of the forward face		0.65%	3.1	Gage
203	Radii or chamfers, missing or incorrect		0.65%	3.1	Visual
204	Foreign matter		0.65%	3.8.5	Visual
205	Burr		0.65%	3.8.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48423A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.4.2	Base, Prior to Coating (as applicable)				8595383
					NEXT HIGHER ASSEMBLY
					8595386
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	True position of collar groove with outside diameter of flange		0.40%	3.1	Gage
102	True position diameter forward of flange with outside diameter of flange		0.40%	3.1	Gage
103	True position primer hole with outside diameter of flange		0.40%	3.1	Gage
104	True position primer hole counterbore with primer hole		0.40%	3.1	Gage
105	Metal defective		0.40%	3.1	Visual
Minor					
201	Diameter of flange, min.		0.65%	3.1	Gage
202	Diameter forward of flange		0.65%	3.1	Gage
203	Diameter of collar groove		0.65%	3.1	Gage
204	Depth of collar groove		0.65%	3.1	Gage
205	Depth of recess in body seat		0.65%	3.1	Gage
206	Width of collar groove		0.65%	3.1	Gage
207	Width of flat in body seat		0.65%	3.1	Gage
208	Thickness of flange min.		0.65%	3.1	Gage
209	Thickness through base below raised surface		0.65%	3.1	Gage
210	Angle of collar groove sidewall incorrect		0.65%	3.1	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48423A (AR)

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
4.4.4.2	Base, Prior to Coating		8595383
			NEXT HIGHER ASSEMBLY
			8595386

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
211	Diameter of raised surface (if applicable)		0.65%	3.1	Gage
212	Thickness through base at raised surface (if applicable)		0.65%	3.1	Gage
213	Surface finish, improper		0.65%	3.1	Visual
214	Length of collar groove protrussion, max. (if applicable)		0.65%	3.1	Gage
215	Width of collar groove protrusion, max. (if applicable)		0.65%	3.1	Gage
216	Maximum offset of the diameter of the outside surface of the collar groove maximum offset of the diameter of the inside surface of the collar groove		0.65%	3.1	Gage
217	Radii or chamfers, missing or incorrect		0.65%	3.1	Visual
218	Height of raised surface in groove, max.		0.65%	3.1	Gage
219	Burr		0.65%	3.8.4	Visual
220	Foreign matter		0.65%	3.8.5	Visual

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.4.2.1	Base, Prior to Coating (as applicable)				11820670
					NEXT HIGHER ASSEMBLY
					8595386
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	True position of body groove diameter with outside diameter of flange		0.40%	3.1	Gage
102	True position small diameter forward of flange with outside diameter of flange		0.40%	3.1	Gage
103	True position primer hole with outside diameter of flange		0.40%	3.1	Gage
104	True position primer hole counterbore with primer hole		0.40%	3.1	Gage
105	Metal defective		0.40%	3.1	Visual
Minor					
201	Diameter of flange min.		0.65%	3.1	Gage
202	Small diameter forward of flange		0.65%	3.1	Gage
203	Diameter of body groove flange		0.65%	3.1	Gage
204	Depth of body seat		0.65%	3.1	Gage
205	Width of flat in body seat		0.65%	3.1	Gage
206	Thickness of flange, min.		0.65%	3.1	Gage
207	Thickness through base at raised surface (if applicable)		0.65%	3.1	Gage
208	Diameter of raised surface max (if applicable)		0.65%	3.1	Gage
209	Thickness through base below raised surface		0.65%	3.1	Gage

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48423A (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.4.2.1	Base, Prior to Coating (as applicable)				11820670
					NEXT HIGHER ASSEMBLY
					8595386
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
210	Length, total		0.65%	3.1	Gage
211	Length of body seat flange		0.65%	3.1	Gage
212	Wall thickness body crimping collar		0.65%	3.1	Gage
213	Surface finish improper		0.65%	3.1	Visual
214	Radii or chamfers, missing or incorrect		0.65%	3.1	Visual
215	Burr		0.65%	3.8.4	Visual
216	Foreign matter		0.65%	3.8.5	Visual

NOTES:

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MIL-C-48423A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.4.2.2	Base, Prior to Coating (as applicable)				11751170
					NEXT HIGHER ASSEMBLY
					8595306
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	True position of body groove diameter with outside diameter of flange		0.40%	3.1	Gage
102	True position diameter forward of flange with outside diameter of flange		0.40%	3.1	Gage
103	True position primer hole with outside diameter of flange		0.40%	3.1	Gage
104	True position primer hole counterbore with primer hole		0.40%	3.1	Gage
105	Metal defective		0.40%	3.1	Visual
Minor					
201	Diameter of flange, min.		0.65%	3.1	Gage
202	Diameter forward of flange		0.65%	3.1	Gage
203	Diameter of body groove flange		0.65%	3.1	Gage
204	Inside diameter of body groove flange recess		0.65%	3.1	Gage
205	Depth of body groove		0.65%	3.1	Gage
206	Depth of body seat recess		0.65%	3.1	Gage
207	Width of body groove		0.65%	3.1	Gage
208	Thickness of flange		0.65%	3.1	Gage
209	Length through base		0.65%	3.1	Gage
210	Length from rear surface to body sidewall seat		0.65%	3.1	Gage

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48423A (AR)

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
4.4.4.2.2	Base, Prior to Coating		11751170
			NEXT HIGHER ASSEMBLY
			8595386

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
211	Length, total		0.65%	3.1	Gage
212	Surface finish improper		0.65%	3.1	Visual
213	Radii or chamfers missing or improper		0.65%	3.1	Visual
214	Burr		0.65%	3.8.4	Visual
215	Foreign matter		0.65%	3.8.5	Visual

NOTES:

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MIL-C-48423A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.4.3	Case, Cartridge, 105MM, M14B4 Assembly				8595386-1-2-3
					NEXT HIGHER ASSEMBLY
					As Applicable
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inside diameter of the mouth, min.		100%	3.1	Gage
102	Maximum profile and alignment		100%	3.1	Gage (4.5.2)
103	Diameter of primer hole		0.40%	3.1	Gage
104	Depth of primer hole counterbore, min.		0.40%	3.1	Gage
105	Depth from front face of shoulder to bottom of primer hole counterbore, min.		0.40%	3.1	Gage
106	Taper in primer hole, incorrect		0.40%	3.1	Gage
107	Collar cracked in weld area (if applicable)		0.40%	3.1	Visual
108	Hardness	10		3.4	4.5.1
109	Number of wraps		0.40%	3.1	Visual
110	Thickness of blanked steel		0.40%	3.1	Gage
111	Body/collar/ base interface	(a)		3.1/ 3.8.3	Visual/4.5.5
Minor					
201	Diameter of primer hole counterbore		0.65%	3.1	Gage
202	Length to forward datum diameter, min.		0.65%	3.1	Gage
203	Length to rear datum diameter, min.		0.65%	3.1	Gage
204	Length total		0.65%	3.1	Gage
205	Protective coating, inadequate or damaged		0.65%	3.2	Visual/4.5.3
206	Overlap at mouth(to be checked after final assembly)		0.65%	3.1	Gage

NOTES: (a) Two per shift per para 4.4.5.3 and 4.5.4.4.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48423A (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.4.3	Case, Cartridge 105MM M14B4 Assembly				8595386-1-2-3
					NEXT HIGHER ASSEMBLY: As Applicable
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
207	Varnish in primer hole		0.65%	3.1	Visual
208	Compound missing in primer hole		0.65%	3.1	Visual
209	Marking missing, incorrect or illegible		0.65%	3.1	Visual
210	Burr		0.65%	3.8.4	Visual
211	Foreign matter		0.65%	3.8.5	Visual
212	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.4.4	Packing, Carton, Prior to Sealing				7548320
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Improper assembly		0.65%	3.1	Visual
202	Incorrect number of cartridge cases in carton		0.65%	3.1	Visual
203	Separation missing or damaged		0.65%	3.1	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48423A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.4.5	Carton, Packed and Sealed				7548320
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Carton damaged, cut or gouged through all layers		0.65%	3.1	Visual
202	Carton improperly sealed		0.65%	3.1	Visual
203	Contents loose		0.65%	3.1	Visual/Manual
204	Marking missing, incorrect or illegible		0.65%	3.1	Visual

NOTES:

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4.4.5 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.5.

4.4.5.1 Protective coating.-The samples to be used for the specified tests shall be coated components, for example base, collar and body. The sample size and acceptance criteria shall be as specified in MIL-C-46487. Failure to comply with the requirements shall be cause for taking the corrective action prescribed herein

4.4.5.2 Functional.-The following proving ground functional tests will be performed by the Government.

4.4.5.2.1 Production lot sample.-One hundred cartridge cases from each production lot of 25,000 or less, will be ballistically tested in the excess pressure phase. When lots exceed 25,000 cartridge cases, 4 additional cartridge cases will be tested for every 1,000 or fraction thereof above 25,000. Acceptance and rejection criteria shall be in accordance with Table II. The lot shall be considered eligible for retest using double the number of samples provided that after firing no cartridge case contains cracks or splits within one caliber of the head, and that the rejection criteria of Table II has not been attained.

TABLE II

Defect	Characteristic	First Sample Accept	First Sample Reject	Cumulative Sample Accept	Cumulative Sample Reject
1.	Obturation	0	2	1	2
2.	Splits or cracks through all layers within 1 caliber of head	0	1	not permitted	
3.	Splits or cracks in sidewall more than 1 caliber from head*	1	3	2	3
4.	Ejection	0	2	1	2
5.	Primer retention	0	2	1	2
6.	Loose assembly	0	1	not permitted	

*Splits or cracks within one caliber of the head that do not extend through all layers of metal shall be considered the same as defect characteristic 3.

4.4.5.3 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.4.1.

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The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of the MIL-A-48078 and 6.6 herein.

4.4.5.4 Body/Base/Collar interface.-In order to determine the integrity of the body/collar crimp two samples shall be inspected from each shifts production as per 4.5.5. The samples shall be one from the middle of the work shift and one from the end of the work shift. An additional sample shall be taken, if any change to the process is made, directly after the change is made. Failure of any sample to comply with the requirements shall be cause for rejection of the product produced subsequent to the last inspection.

4.5 Test methods and procedures

4.5.1 Hardness.-Samples selected in accordance with 4.4.4.3 will be tested as follows: The body shall be removed from the base by cutting through the body approximately one inch from the rear face of the base. The bottom portion of the body and the collar shall be removed from the base. Seven hardness values shall be taken across the center of the rear face of the base on both sides of the primer hole at approximately 1/4 inch intervals. Four hardness values shall be taken on the underside of the flange of the collar approximately 90 degrees apart. The large portion of the body shall be hand straightened to permit hardness values to be taken, both vertically and horizontally at approximately 2 inch intervals, through the center of the length and width of the piece. All hardness values shall be taken on bare metal, and in accordance with the procedures prescribed by ASTM E-18.

4.5.2 Maximum profile and alignment gage.-The cartridge case, after application of the protective coating shall freely enter the maximum profile gage described on drawing 8647259. The gate of the gage shall close without interference with the cartridge case head.

4.5.3 Protective coating.-The methods and procedures for performing these tests will be in accordance with the requirements of MIL-C-46487.

4.5.3.1 Salt spray.-The salt spray test shall be performed using a 20 percent sodium chloride solution with the equipment and criteria prescribed by ASTM Method B117

4.5.4 Functional

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4.5.4.1 High pressure application (See paragraph 6.9)

4.5.4.1.1 Excess, excess pressure phase.-The cartridge case shall be assembled into a complete round in the same manner as required for service firing. The propellant charge will be such as to produce a chamber pressure of 52,100 psi (359.2 MPa) to 54,700 psi (377.1 MPa). The complete round will be temperature conditioned at plus 70°F ± 5° (21.1°C ± 2.8°) and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

4.5.4.1.2 Excess pressure phase.-The cartridge case shall be assembled into a complete round in the same manner as required for service firing. The propellant charge will be such as to produce a min. mean chamber pressure of 48,400 psi (333.7 MPa) to max. individual pressure of 52,100 psi (359.2 MPa). The complete round will be conditioned at plus 70°F ± 5° (21.1°C ± 2.8°) and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

4.5.4.1.3 Cold temperature phase.-The cartridge case shall be assembled into a complete round in the same manner as required for service firing. The propellant charge will be such as to produce a chamber pressure of 42,000 psi (289.6 MPa) to 44,000 (303.4 MPa). The complete round will be temperature conditioned at minus 35°F (-37.2°C) to minus 45°F (-42.8°C) and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

4.5.4.2 Standard application (See paragraph 6.9)

4.5.4.2.1 Excess, excess pressure phase.-The cartridge case shall be assembled into a complete round in the same manner as required for service firing. The propellant charge will be such as to produce a chamber pressure of 45,600 psi (314.4 MPa) to 46,800 psi (322.7 MPa). The complete round will be temperature conditioned at plus 70°F ± 5° (21.1°C ± 2.8°) and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

4.5.4.2.2 Excess pressure phase.-The cartridge case shall be assembled into a complete round in the same manner as required for service firing. The propellant charge will be such as to produce a minimum mean chamber pressure of 42,500

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psi (293.0 MPa) to maximum individual pressure of 45,600 psi (314.4 MPa). The complete round will be conditioned at plus 70°F ± 5° (21.1°C ± 2.8°) and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

4.5.4.2.3 Cold temperature phase.-The cartridge case shall be assembled into a complete round in the same manner as required for service firing. The propellant charge will be such as to produce a chamber pressure of 35,000 psi (241.3 MPa) to 38,000 (262.0 MPa). The complete round will be temperature conditioned at minus 35°F (-37.2°C) to minus 45°F (-42.8°C) and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

4.5.5 Body/Base/Collar interface.-The samples taken in accordance with 4.4.5.4 shall be cut along a center line of the cartridge case and inspected for the following characteristics:

a.	Distance from outside wall of base to a single ply of the body	Gage
b.	Metal defects	Visual (see 3.8.2)
c.	Poor workmanship	Visual (see 3.8)

All inspections shall be on the same sectioned part and shall be gaged on the same half of the part.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packing

5.1.1 Level C-minimum military pack.-The cartridge cases shall be packed in accordance with requirements of drawing 7548320.

5.2 Labeling and marking.-Labeling and marking for shipment shall be in accordance with MIL-STD-1169.

5.3 Data card.-Data card information shall be in accordance with MIL-STD-1167.

6. NOTES

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6.1 Prior to fabrication of a first article sample the contractor upon approval of the contracting officer may make and submit for evaluation up to five (5) process control lots. These process control lots shall consist of not less than ten (10) nor more than twenty-five (25) cartridge cases each. The contractor shall furnish all information pertaining to the manufacture of the lots to the contracting officer. All costs for testing these process control lots will be borne by the Government.

6.2 The term "ejected" shall mean that the cartridge case is thrown free of the breech of the weapon to permit loading of the subsequent round.

6.3 Visual examination qualification.-When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.4 Ordering data.-See MIL-A-48078.

6.5 Mechanical properties.-The number of specimens and their location are for the convenience of the contractor and government and shall not restrict the contractor or government from testing additional specimens at other locations to determine that properties of the components conform to drawing requirements.

6.6 Submission of inspection equipment designs for approval.-See MIL-A-48078.-Submit equipment designs, as required to:

Commander
US Army Armament Research and Development Command
ATTN: DRDAR-QAR-I
Dover, NJ 07801

6.7 Drawing.-Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

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6.8 Submission of test data and ammunition data cards.-In addition to the normal distribution of records when the projectile is procured by the Army Armament Material Readiness Command, one copy of all ballistic test data and ammunition data cards shall be forwarded to each of the following: Commander, ARRADCOM, ATTN: DRDAR-LCU-M and DRDAR-QAR-E, Dover, N.J. 07801; and Commander, ARRCOM, ATTN: DRSAR-QAM, Rock Island, ILL. 61299.

6.9 The M14B4 cartridge case has been designed, tested and type classified to operate at the pressure levels of paragraph 4.5.4.1. These pressure levels are required for use with extended range, high pressure gun systems i.e M204, which are not currently available. Currently available gun systems operate at the pressure levels of paragraph 4.5.4.2. Cases intended for use with extended range, high pressure gun systems i.e M204, shall be tested at the pressure levels of paragraph 4.5.4.1.

Cases intended for use with the standard M101A1 and M102 systems may be tested in accordance with paragraph 4.5.4.2 provided ammunition lot data cards and firing records are annotated as follows: "Case, cartridge, M14B4 is not to be used for extended range high pressure rounds unless retested ballistically for such pressures."

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Army-AR

Preparing Activity:
Army-AR

Project No.: 1315-A202

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-48423A Case, Cartridge, 105MM, M14B4

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☐ VENDOR ☐ USER ☐ MANUFACTURER

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